

Review of Last Lecture

1.5 Decomposition of Signals

1.6 System Models and Classification

1.7 Linear Time-Invariant Systems

1.8 System Analysis Methods

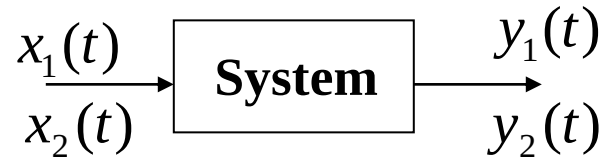
1.7 Linear Time-Invariant Systems



1.7.1 Linearity

Linearity includes superposition and homogeneity (also called scalability).

1. Superposition



If $x(t) = x_1(t) + x_2(t)$, $y(t) = y_1(t) + y_2(t)$,
the system is said to satisfy superposition.

2. Homogeneity

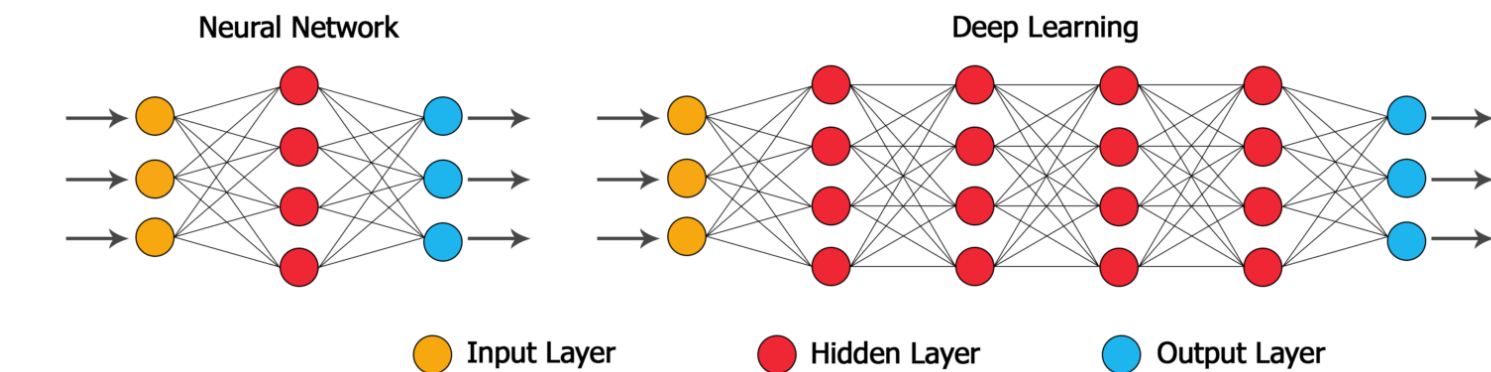
If $x(t) = ax_i(t)$, $y(t) = ay_i(t)$,

the system is said to satisfy homogeneity.

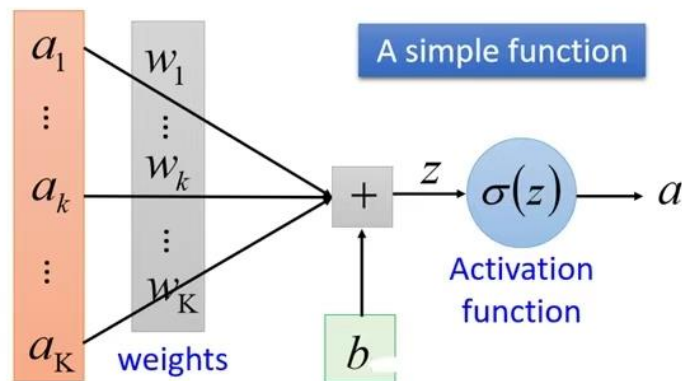
A system that satisfies both superposition and homogeneity is called a **linear system**.

Consider a deep neural network. Assume the sigmoid function $f(x) = \frac{1}{1+e^{-x}}$ is used for activation. Determine whether it is a linear system. ()

- ☐ A Linear
- ☒ B Nonlinear



$$z = a_1 w_1 + \dots + a_k w_k + \dots + a_K w_K + b$$



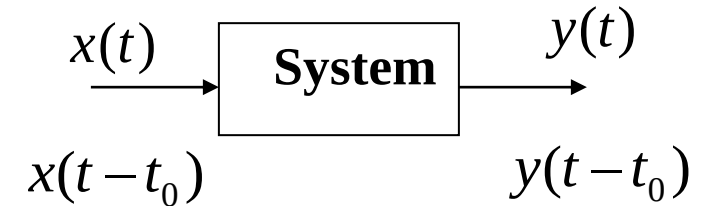
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1.7 Linear Time-Invariant Systems



1.7.2 Time-Invariance

Time-invariant system: The parameters of the system do not change with time; if the excitation is delayed, the response is delayed by the same amount.



To determine whether a system satisfies a certain property, **we only need to find one example that does not satisfy it** to prove that the system does not have that property.

Intuitive Judgment Method for Time-Invariance:

If there is a time-varying coefficient in front of $x(\cdot)$ (e.g., $tx(t)$), or reverse (e.g., $x(-t)$), or scaling transformation (e.g., $x(2t)$), then the system is time-varying.

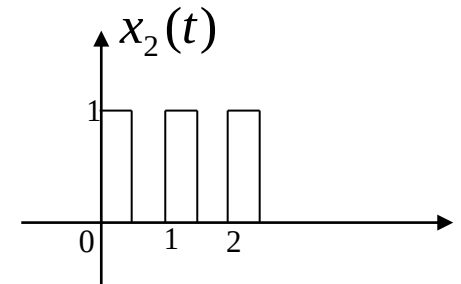
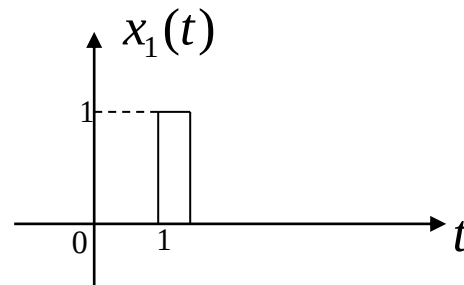
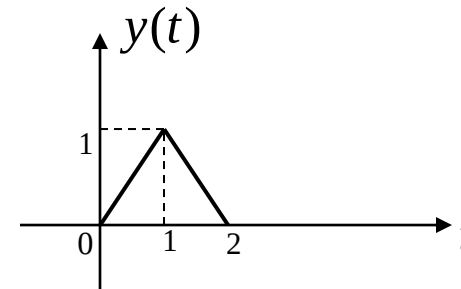
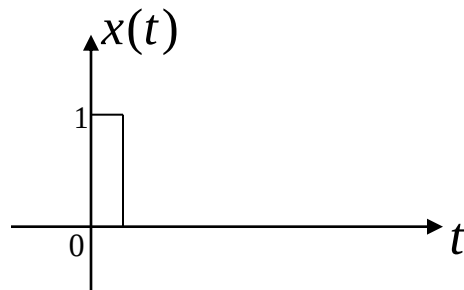
1.7 Linear Time-Invariant Systems



A system that satisfies both linearity and time-invariance is called a **Linear Time-Invariant (LTI)** system, which can be expressed as:

$$x(t) = \sum_{k=1}^N a_k x_k(t - t_k) \quad y(t) = \sum_{k=1}^N a_k y_k(t - t_k)$$

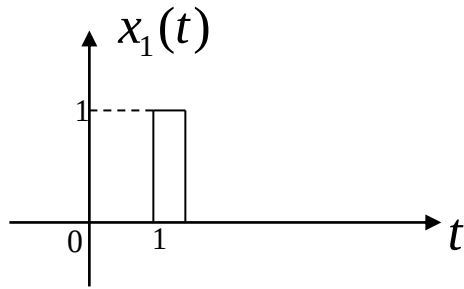
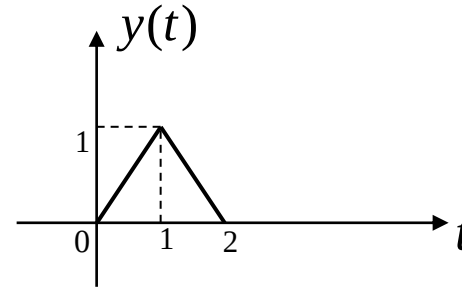
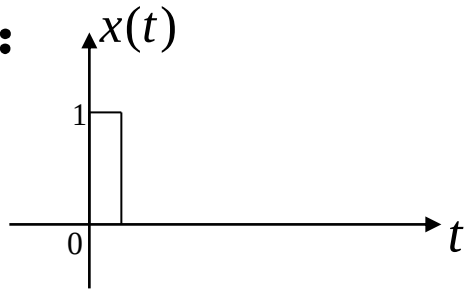
Example 1-15: Assume the input-output relationship of an LTI system is described in the figure. Draw the waveforms of outputs $y_1(t)$ and $y_2(t)$ when the inputs are $x_1(t)$ and $x_2(t)$ respectively.



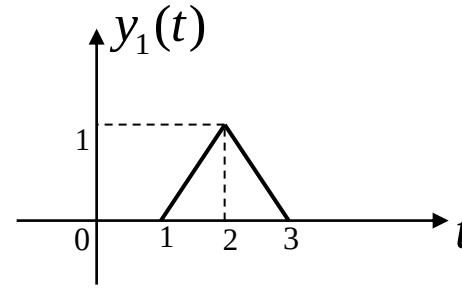
1.7 Linear Time-Invariant Systems



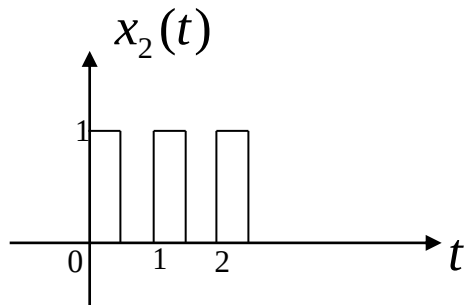
Solution:



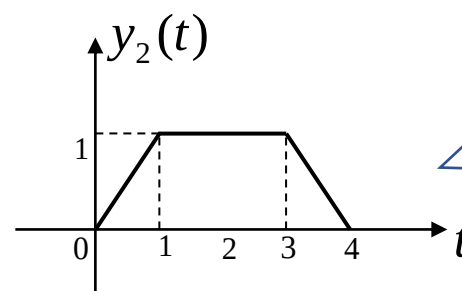
$$x_1(t) = x(t-1)$$



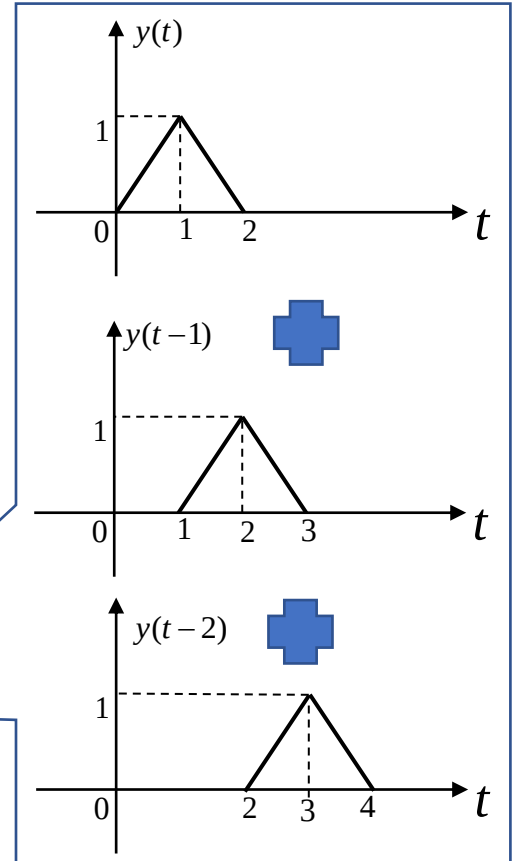
$$y_1(t) = y(t-1)$$



$$x_2(t) = x(t) + x(t-1) + x(t-2)$$



$$y_2(t) = y(t) + y(t-1) + y(t-2)$$



1.7.3 Causality

A causal system is a system where the response at time t_0 depends only on the input at time $t \leq t_0$. Otherwise, it is a non-causal system.

Examples:

Causal system: $r(t) = e(t - 1)$ (delay system)

Non-causal system: $r(t) = e(t + 1)$ (advance system)

(The response at time $t = 0$ is determined by the excitation at time $t = 1$, so it is a non-causal system.)

Non-causal system: $r(t) = e(2t)$ (time-compressed system)

Non-causal systems are common in fields such as speech signal processing, meteorology, stock market analysis, and demography.

1.8 Analysis Methods of LTI Systems



Continuous –Time



Discrete –Time



Time-Domain Analysis

Transform-Domain Analysis

Frequency-Domain
Complex Frequency-Domain

State-Space Analysis

Engineering Applications

Chapter Contents

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**2.2 Establishment of System Mathematical Model
(Differential Equation)**

**2.3 Solving Differential Equations Using the
Classical Time-Domain Method**

**2.4 Jump at the Starting Point - Conversion from
 0_- to 0_+ State**

2.5 Zero-Input Response and Zero-State Response

2.6 Impulse Response and Step Response

2.7 Convolution

2.8 Properties of Convolution

Contents of This Lecture

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2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

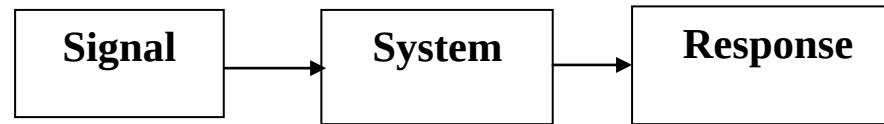
Learning Objectives of This Lecture

1. Familiarize with the method of establishing system differential equations (how to eliminate intermediate variables);
2. Proficiently master the classical method to solve the **free response** and **forced response** of the system;
3. Proficiently use the **impulse function matching method** to determine the jump variable of the starting point state.

Why Time-Domain Analysis of Systems?

Time-domain analysis of LTI systems offers:

- Intuitive physical insights, serving as the foundation for transform-domain analysis methods;
- Complex manual calculations, but software/tools (e.g., MATLAB) simplify solving.



Core Questions Addressed

For a given signal, a defined system, and the goal of finding the system response:

- How to establish the mathematical model of the system? (**Differential equations**)
- How to decompose the system response? (**Free response + Forced response; Zero-input response + Zero-state response**)
- How to solve different types of responses? (**Classical method & convolution method**)
- How to apply signals and systems? (**Simulation diagrams & computer-aided solving**)

2.2 Establishment of System Mathematical Model (Differential Equation)

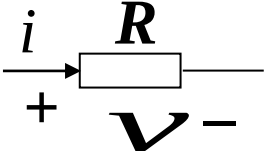


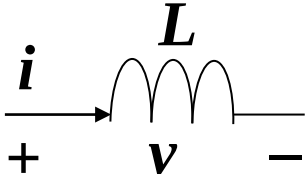
The goal of establishing a differential equation is to mathematically describe the physical characteristics of a system.

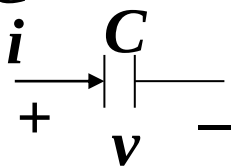
Electrical Circuit System Model

1) Component Constraints (Circuit Elements)

Voltage-Current Relationship

Resistor R  $v = Ri$

Inductor L  $i = \frac{1}{L} \int v dt$ $v = L di / dt$

Capacitor C  $i = C \frac{dv}{dt}$ $v = \frac{1}{C} \int i dt$

2) Network Topology Constraints

Kirchhoff's Current Law (KCL): At any instant, the sum of currents entering a node equals the sum of currents leaving the node:

$$\sum I_{in} = \sum I_{out}$$

Kirchhoff's Voltage Law (KVL): Around any closed loop, the sum of electromotive forces equals the sum of voltage drops across resistors:

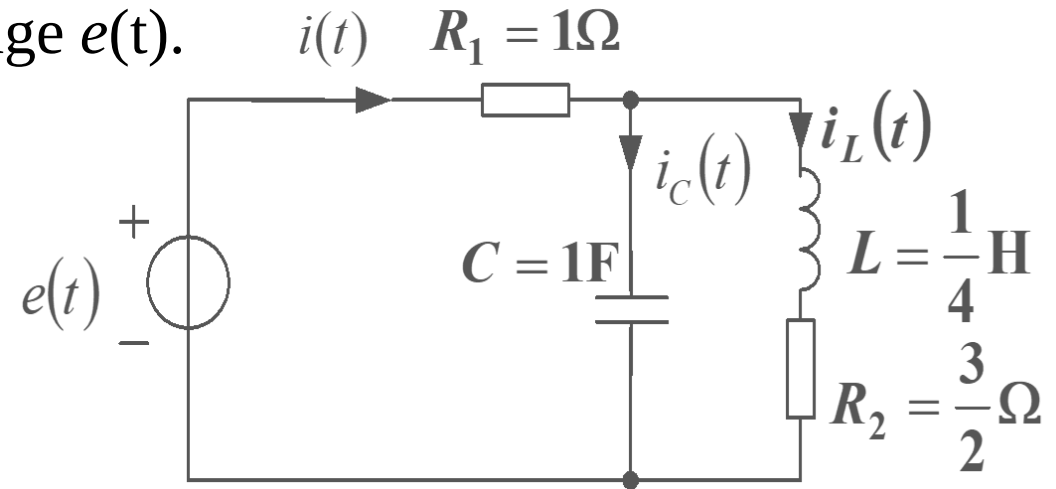
$$\sum_k E_k = \sum_k U_k$$

2.2 Establishment of System Mathematical Model (Differential Equation)



Establishing a Differential Equation for Circuit Current

Example 2-1: For the given circuit (as shown in the figure), establish the differential equation for current $i(t)$ with input voltage $e(t)$.



Solution:

List KVL/KCL equations:

Loop voltage equation: $R_1 i(t) + v_C(t) = e(t)$ (1)

Loop voltage equation: $v_C(t) = L \frac{d}{dt} i_L(t) + i_L(t) R_2$ (2)

Node current equation: $i(t) = C \frac{d}{dt} v_C(t) + i_L(t)$ (3)

2.2 Establishment of System Mathematical Model (Differential Equation)



Eliminate intermediate variables, i.e., $v_C(t)$ and $i_L(t)$.

$$R_1 i(t) + v_C(t) = e(t) \quad (1)$$

(1) is simplest, containing $v_C(t)$ only, not $v_C'(t)$.

$$v_C(t) = L \frac{d}{dt} i_L(t) + i_L(t) R_2 \quad (2)$$

(2) is most complex, containing $i_L'(t)$ and related to $v_C''(t)$.

$$i(t) = C \frac{d}{dt} v_C(t) + i_L(t) \quad (3)$$

(3) Contains $v_C'(t)$.

From (1), derive $v_C(t) = e(t) - R_1 i(t) \quad (4)$

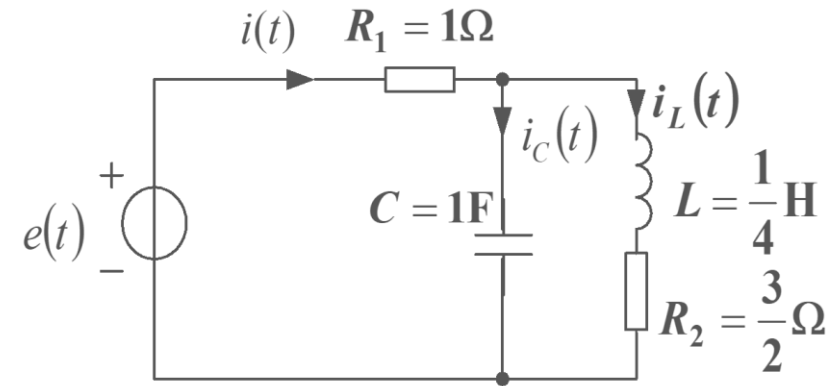
Substituting (4) into (3), derive $i_L(t) = i(t) + CR_1 \frac{d}{dt} i(t) - C \frac{d}{dt} e(t) \quad (5)$

Substitute (4) and (5) into (2), resulting in a single differential equation for $i(t)$ and $e(t)$ ((6) on next page).

2.2 Establishment of System Mathematical Model (Differential Equation)



$$\begin{aligned} & \frac{d^2}{dt^2} i(t) + \left(\frac{1}{R_1 C} + \frac{R_2}{L} \right) \frac{d}{dt} i(t) + \left(\frac{1}{LC} + \frac{R_2}{R_1 LC} \right) i(t) \\ &= \frac{1}{R_1} \frac{d^2}{dt^2} e(t) + \frac{R_2}{R_1 L} \frac{d}{dt} e(t) + \frac{1}{R_1 LC} e(t) \end{aligned} \quad (6)$$



Substituting the component parameters into (6) yields

$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10 i(t) = \frac{d^2}{dt^2} e(t) + 6 \frac{d}{dt} e(t) + 4 e(t)$$

For complex systems, the relationship between the input $x(t)$ and output $y(t)$ is described by an **n -th order linear constant-coefficient ordinary differential equation (ODE)**:

$$\begin{aligned} a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) \\ = b_m \frac{d^m x(t)}{dt^m} + b_{m-1} \frac{d^{m-1} x(t)}{dt^{m-1}} + \dots + b_1 \frac{dx(t)}{dt} + b_0 x(t) \end{aligned}$$

The **complete solution** of this ODE consists of two parts:

- 1) **Homogeneous solution**: Also called **free response**, it describes the system's inherent behavior (determined by the system itself).
- 2) **Particular solution**: Also called **forced response**, it describes the system's behavior under the influence of the input (determined by the input signal).

2.3 Solving Differential Equations by the Classical Time-Domain Method

Step 1: Solve the Homogeneous Solution

The homogeneous solution satisfies the **homogeneous ODE** (set the input side to 0):

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = 0$$

Characteristic Equation: $a_n \alpha^n + a_{n-1} \alpha^{n-1} + \dots + a_1 \alpha + a_0 = 0$

The roots $\alpha_1 \dots \alpha_n$ are called the system's “natural frequencies”.

Roots of the Characteristic Equation	Form of Homogeneous Solution (Coefficients $A_1, \dots, A_n$ are undetermined)
Distinct real roots	$y_h(t) = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t} + \dots + A_n e^{\alpha_n t}$
Repeated real roots	If α_1 is a k -th order repeated root, the corresponding term is $(A_1 t^{K-1} + A_2 t^{K-2} + \dots + A_{K-1} t + A_K) e^{\alpha_1 t}$
Complex conjugate roots	If $\alpha_{1,2} = \alpha \pm j\beta$, the corresponding term is $e^{\alpha t} (A_1 \cos \beta t + A_2 \sin \beta t)$

2.3 Solving Differential Equations by the Classical Time-Domain Method

Step 2: Solve the Particular Solution

The form of the particular solution depends on the input signal (or the “**free term**” of the ODE, i.e., the right-hand side of the ODE).

Free Term	Form of Particular Solution $y_p(t)$
E (constant)	B (constant)
t^p	$B_p t^p + B_{p-1} t^{p-1} + \dots + B_1 t + B_0$
$e^{\alpha t}$	$\begin{cases} Be^{\alpha t} & (\text{If } \alpha \text{ is not a characteristic root.}) \\ Bt^k e^{\alpha t} & (k: \text{multiplicity of } \alpha) \end{cases}$
$\cos \omega_0 t / \sin \omega_0 t$	$B_1 \cos \omega_0 t + B_2 \sin \omega_0 t$

Step 3: Find Undetermined Coefficients

Substitute the complete solution $\mathbf{y}(t) = \mathbf{y}_h(t) + \mathbf{y}_p(t)$ into the original ODE, and solve for the undetermined coefficients A_1, A_2, A_n using the initial conditions (0_+ states).

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Definition of States

1. **0_- state**: The system state just before the input is applied (no input influence).
2. **0_+ state**: The system state just after the input is applied (used as initial conditions for solving the ODE, since **the ODE solution is valid for $t \geq 0_+$**).

Circuit Switching Rule

For circuit systems, the 0_- state describes the energy storage of components (capacitor voltage $v_C(0_-)$, inductor current $i_L(0_-)$).

In most cases, **the voltage across a capacitor and the current through an inductor do not change abruptly during switching**: $v_C(0_-) = v_C(0_+)$, $i_L(0_-) = i_L(0_+)$

However, **if an impulse current is applied to a capacitor or an impulse voltage to an inductor, $i_C(t)$ or $v_L(t)$ will jump between 0_- and 0_+** .

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

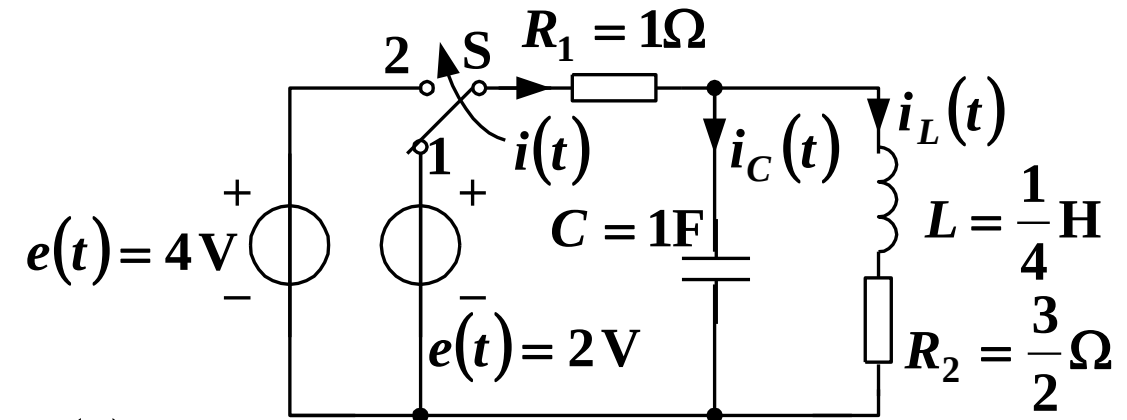
Example 2-2: Solve the differential equation established in Example 2-1. Assume that the switch moves from 1 to 2 when $t=0$.

Solution:

The differential equation established is

$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10i(t) = \frac{d^2}{dt^2} e(t) + 6 \frac{d}{dt} e(t) + 4e(t)$$

The input signal can be expressed as $e(t) = 2u(t) + 2 \text{ V}$.



Step 1: Find the homogeneous solution

- 1) Characteristic equation: $\alpha^2 + 7\alpha + 10 = 0 \rightarrow$ Roots $\alpha_1 = -2, \alpha_2 = -5$;
- 2) Homogeneous solution: $i_h(t) = A_1 e^{-2t} + A_2 e^{-5t}$, where A_1 and A_2 are undetermined.

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 2: Find the particular solution

- 1) Substitute the input $e(t)=2u(t)+2$ in the right side of the differential equation and **keep only the nonzero terms when $t>0$,**

$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10i(t) = \frac{d^2}{dt^2} e(t) + 6 \frac{d}{dt} e(t) + 4e(t)$$



$$\frac{d^2}{dt^2} i(t) + 7 \frac{d}{dt} i(t) + 10i(t) = 16$$

- 2) Assume $i_p(t) = B$;
- 3) Substitute $i_p(t)=B$ into the ODE, $10B=16 \rightarrow B = 1.6$.

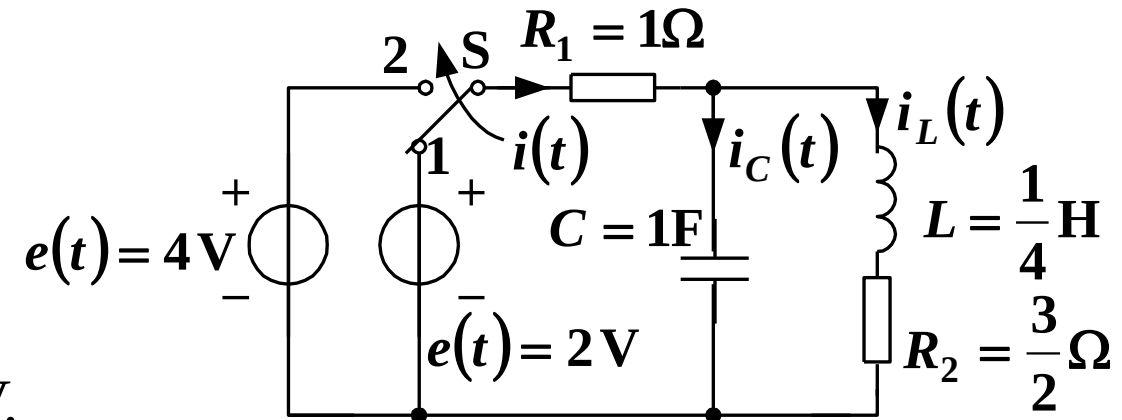
2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 3: Complete solution

$$i(t) = i_h(t) + i_p(t) = A_1 e^{-2t} + A_2 e^{-5t} + 1.6$$

When $t=0_-$, $i(0_-) = e(0_-) / (R_1 + R_2) = 0.8$ A,

$$i_L(0_-) = i(0_-) = 0.8 \text{ A}, \quad V_C(0_-) = i_L(0_-) R_2 = 1.2 \text{ V}.$$



As the voltage across a capacitor and the current through an inductor do not change from $t=0_-$ to 0_+ , $V_C(0_+) = V_C(0_-) = 1.2$ V, $i_L(0_+) = i_L(0_-) = 0.8$ A.

$$i(0_+) = \frac{1}{R_1} [e(0_+) - v_C(0_+)] = 4 - 1.2 = 2.8 \text{ A}$$

$$\begin{aligned} \frac{d}{dt} i(0_+) &= \frac{1}{R_1} \left[\frac{d}{dt} e(0_+) - \frac{d}{dt} v_C(0_+) \right] = \frac{1}{R_1} \left[0 - \frac{i_C(0_+)}{C} \right] \\ &= \frac{i_L(0_+) - i(0_+)}{R_1 C} = i_L(0_-) - i(0_+) = 0.8 - 2.8 = -2 \text{ A/s} \end{aligned}$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 4: Determine Coefficients A_1 and A_2

$$As \quad i(t) = A_1 e^{-2t} + A_2 e^{-5t} + 1.6 \quad \frac{d}{dt} i(t) = -2A_1 e^{-2t} - 5A_2 e^{-5t}$$

$$\begin{cases} i(0_+) = A_1 + A_2 + 1.6 = 2.8 \\ \frac{d}{dt} i(0_+) = -2A_1 - 5A_2 = -2 \end{cases} \quad \rightarrow \quad \begin{cases} A_1 = \frac{4}{3} \\ A_2 = -\frac{2}{15} \end{cases}$$

The complete solution is

$$i(t) = \left(\frac{4}{3} e^{-2t} - \frac{2}{15} e^{-5t} + \frac{8}{5} \right) A \quad (t > 0)$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Example 2-3: For the given circuit (as shown in the figure), input $x(t)=\cos(2t)u(t)$, initial voltages of both capacitors are zero. Find the response $v_2(t)$.

Solution:

Step 1: Establish the ODE

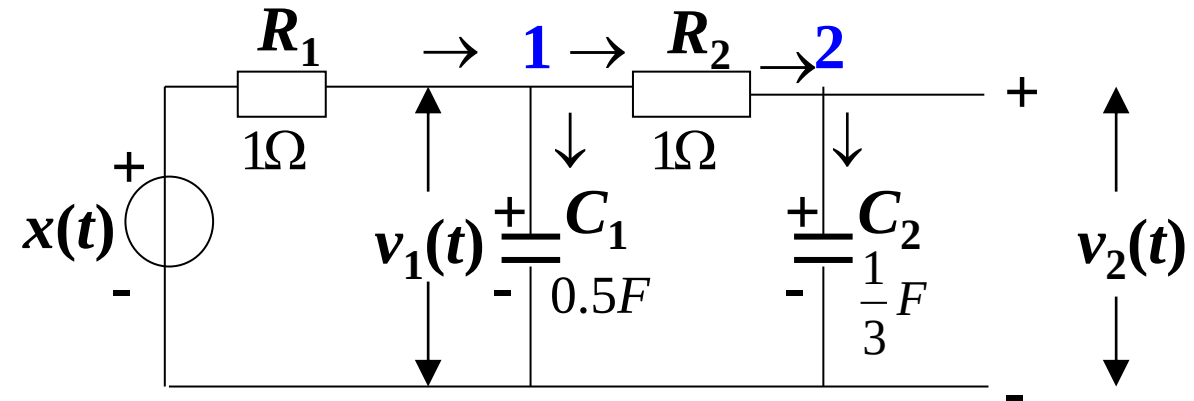
List KCL equations for nodes 1 and 2,

$$\left\{ \begin{array}{l} \frac{x(t) - v_1(t)}{R_1} = C_1 \frac{dv_1(t)}{dt} + \frac{v_1(t) - v_2(t)}{R_2} \end{array} \right. \quad (1)$$

$$\left\{ \begin{array}{l} \frac{v_1(t) - v_2(t)}{R_2} = C_2 \frac{dv_2(t)}{dt} \end{array} \right. \quad (2)$$

From (2), derive $v_1(t) = R_2 C_2 \frac{dv_2(t)}{dt} + v_2(t)$ (3)

Substituting (3) into (1), derive the ODE $\frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6x(t)$



2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

$$\text{ODE: } \frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6x(t)$$

Step 2: Find the homogeneous solution

$$\text{Characteristic equation: } \alpha^2 + 7\alpha + 6 = 0 \quad \text{Roots: } \alpha_1 = -1, \alpha_2 = -6$$

$$\text{Homogeneous solution: } A_1 e^{-t} + A_2 e^{-6t} \quad (A_1 \text{ and } A_2 \text{ are undetermined})$$

Step 3: Find the particular solution

Substitute $x(t) = \cos(2t)$ ($t > 0$) into the right side of the ODE,

$$\frac{d^2 v_2(t)}{dt^2} + 7 \frac{dv_2(t)}{dt} + 6v_2(t) = 6 \cos 2t$$

so assume the particular solution is $B_1 \sin 2t + B_2 \cos 2t$. Substitute it into the left of the ODE,

$$(2B_1 - 14B_2) \sin 2t + (14B_1 + 2B_2) \cos 2t = 6 \cos 2t$$

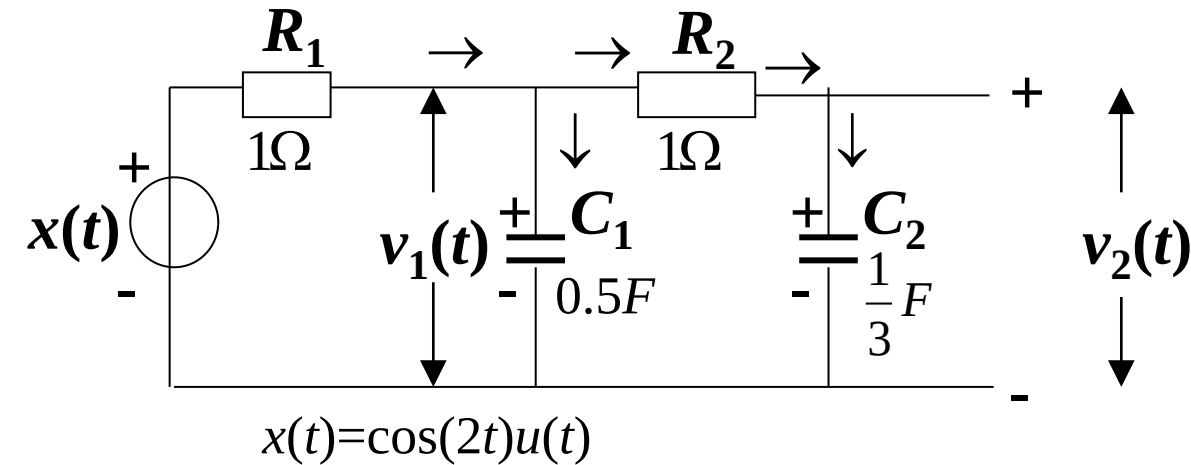
$$\begin{cases} 2B_1 - 14B_2 = 0 \\ 14B_1 + 2B_2 = 6 \end{cases} \longrightarrow B_1 = \frac{21}{50}, B_2 = \frac{3}{50}$$

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Step 4: Find the complete solution

$$v_2(t) = A_1 e^{-t} + A_2 e^{-6t} + \frac{21}{50} \sin 2t + \frac{3}{50} \cos 2t \quad (t > 0)$$

$$\frac{dv_2(t)}{dt} = -A_1 e^{-t} - 6A_2 e^{-6t} + \frac{21}{25} \cos 2t - \frac{3}{25} \sin 2t$$



As $v_2(0_-) = 0$, and the voltage across C_2 does not change abruptly at $t=0$, hence $v_2(0_+) = 0$; Similarly, $v_1(0_-) = v_1(0_+) = 0$. The voltage across R_2 remains at 0 from 0_- to 0_+ , and so there is no current through R_2 and C_2 . Hence, $\frac{dv_2(0_+)}{dt} = 0$.

With $v_2(0_+) = 0$, $A_1 + A_2 = -\frac{3}{50}$

With $\frac{dv_2(0_+)}{dt} = 0$, $A_1 + 6A_2 = \frac{21}{25}$



$$A_1 = -\frac{6}{25}, A_2 = \frac{9}{50}$$

Complete solution: $v_2(t) = -\frac{6}{25} e^{-t} + \frac{9}{50} e^{-6t} + \frac{21}{50} \sin(2t) + \frac{3}{50} \cos(2t) \quad (t > 0)$

- Repeat the examples in the lecture notes.
- 2.52(a) and 2.53(b) in the textbook.